Cancelled Orders Cleanup

Category B Coding Assignment - Problem 2

Problem summary

Remove every node equal to a requested code and return the possibly new head. The algorithm uses a custom node and does not create replacement nodes.

Java solution

```
public class Main {
 static class Node { int data; Node next; Node(int d){data=d;} }
 static Node removeAll(Node head,int code){
  while(head!=null && head.data==code)head=head.next;
  Node current=head;
  while(current!=null && current.next!=null){
   if(current.next.data==code)current.next=current.next.next;
   else current=current.next;
  }
  return head;
 }
 static Node build(int... a){Node h=null,t=null;for(int x:a){Node n=new Node(x);if(h==null)h=n;else t.next=n;t=n
 static void print(Node n){while(n!=null){System.out.print(n.data+" -> ");n=n.next;}System.out.println("null");}
 public static void main(String[] args){print(removeAll(build(5,3,5,8,5),5));print(removeAll(build(5,5,5),5));}
}
```

Sample output

```
3 -> 8 -> null
null
```

Complexity

Time: O(n), each node is processed at most a constant number of times. Additional space: O(1).